State density: integrability, Mellin transform and the leading coefficient (Stage 1c–d)

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Abstract

Units 225–226 complete Stage 1 of the Taylor-tree programme. Unit 225 exports the exact state density $v = \mathrm{eval}(\mathrm{stateDensityRep}\, n\, w)$ of unit 224 in the forms later stages need: integrability on $(0, 1]$ (from the box identity with $g = 1$, no analysis of individual power–log terms), the real-valued identity $\int_{(0,1]^{n+1}} \prod a_i^{w_i} f(\prod a_i)\, da = \int_0^1 vf$ for measurable $f \geq 0$ (both sides as `toReal` of the $\mathbb{R}^{\infty}_{\geq 0}$ identity, no integrability side conditions), the Mellin transform $\int_0^1 z^s v(z)\, dz = \prod_i 1/(w_i + s + 1)$ (the paper's product zeta function in the weight normalisation, by the same one-coordinate recursion), and the Mellin transform of a basis term $\int_0^1 \tau^{c-1}(-\log \tau)^j\, d\tau = j!/c^{j+1}$ (substitution $\tau = e^{-x}$ into Euler's integral), hence termwise $\int_0^1 z^s \,\mathrm{eval}\, c = \sum c_t\, j_t!/(s + \mu_t)^{j_t+1}$. Unit 226 proves the **top coefficient**: with $l = \min(w_i + 1)$ of multiplicity m,

$$c_{l,m-1} = \frac{1}{(m-1)!} \prod_{w_i+1 \neq l} \frac{1}{w_i + 1 - l},$$

by an Abelian argument identical in spirit to Headline XXI: $(s+l)^m \int_0^1 z^s v$ tends, as $s \to -l^+$, to that product (from the product transform) and to $(m-1)!\, c_{l,m-1}$ (termwise: every term but the top one vanishes in the limit). In the monomial normalisation the coefficient is $a_{-m}/(m-1)!$, $a_{-m} = \mathrm{mellinCoeff}\, h\, k\, l\, 1$ (`monomial_leadCoeff`): Astra #26's factorial conversion between density and Laurent coefficients, as a theorem. Zero `sorry`/`axiom`; 9103 jobs.

1 The things formalised

```
theorem weightedBoxIntegral_rpow : WBI d w (ofReal   (·^s)) =   ofReal (1/(w +s+1))
theorem stateDensity_integrableOn (hw :   i, -1 < w i) : IntegrableOn (eval rep) (Ioc 0 1)
theorem integral_unitBox_eq_stateDensity (hf : Measurable f)
  (hf0 :   z   Ioc 0 1, 0   f z) :
    a in unitBox (n+1), (  a ^{w }) * f (  a ) =   z in Ioc 0 1, eval rep z * f z
theorem mellin_stateDensity :   z in Ioc 0 1, z^s * eval rep z =   1/(w +s+1)
theorem integral_Ioc_rpow_mul_neg_log_pow (hc : 0 < c) :   ^(c-1)(-log  )^j = j!/c^(j+1)
theorem integrableOn_powLogBasis (hc : 0 < c) (j)
theorem mellin_eval (hc :   t   c, 0 < t.1 + s) :   z^s eval c =   c  j !/(s+ )^(j +1)
def PowLogRep.coeffAt c   j;  theorem term_tendsto; list_tendsto; product_tendsto
theorem stateDensityRep_leadCoeff (hl :   i, l   w i + 1) (hatt) :
  coeffAt (stateDensityRep n w) l (expMult w l - 1)
    = (1/(expMult w l - 1)!) *   i, if w i + 1 = l then 1 else 1/(w i + 1 - l)
theorem monomial_leadCoeff : (  1/(2k )) * coeffAt (rep (ratioExp - 1)) l (m-1)
    = mellinCoeff h k l 1 / (m-1)!
```

2 Mathematics

The Abelian limit is the natural tool once the Mellin transform is a product: multiplying by $(s+l)^m$ cancels the m factors at the minimum, and the surviving product is continuous at $s=-l$ because the other exponents are strictly larger. On the density side the degree bound $j \le m-1$ at $\mu = l$ (unit 224) is exactly what makes every lower-degree term vanish: $(s+l)^{m-1-j} \to 0$ for $j < m-1$. No inverse Mellin transform and no partial-fraction formula for the other coefficients are needed for the leading term; the full residue formula (`eq:residuecoefficients`) remains a corollary for later, checked numerically in the unit 224 replica.

3 Lean notes

- 'λ' is the lambda token and cannot name a variable; use 'l'.
- `tendsto_const_nhds.div h hne` needs the constant fixed: `(tendsto_const_nhds (x := (1: ))).div`.
- `rw [show m = a + b by omega]` rewrites every `m`, including inside the exponent being manipulated; state the power identity as a separate `have` instead.
- Real identity for nonnegative f without integrability: `integral_eq_lintegral_of_nonneg_ae` on both sides, then the $\mathbb{R}^{\infty}_{\geq 0}$ theorem.
- The substitution $\tau = e^{-x}$: `integral_image_eq_integral_abs_deriv_smul` on `Ioc 0 1` with `-log`, image `Ici 0`; its `iff` version transports integrability.

4 Status

Stage 1 complete (Headlines XXII, XXII$'$, XXII$''$). Next: review v18 of units 223–226; Stage 2 — the exact monomial moment identity with constant phase: $\int_{(0,1]^d} u^h(\sqrt{n}u^k)^p e^{-\beta n u^{2k}+\beta\sqrt{n}u^k\xi_0}\,du = \sum_{\mu,j}\sum_{k'} n^{-\mu}(\log n)^{k'} c_{\mu,j}\binom{j}{k'}\int_0^n t^{\mu+p/2-1}(-\log t)^{j-k'}e^{-\beta t+\beta\sqrt{t}\xi_0}\,dt$ exactly, then the tail $\int_n^\infty$ exponentially small.